

4.6 A Transitive Solution to the Railway Points Problem

THE preceding sections have developed the theory of transitive relations in an entirely abstract setting. It is therefore perhaps surprising that so purely algebraic a notion should find a natural and fruitful application in the design of railway junctions. The connection was first observed—so far as we have been able to ascertain—in a brief notice published in the *Mathematikai és Fizikai Lapok*¹ for the year 1906. The notice, occupying little more than a page, describes a definite problem: to arrange unidirectional switches at a railway junction so that a train may proceed from any incoming track to any outgoing track *without reversal of direction*, and to determine the least number of switches sufficient for this purpose. The reviewer noted that the problem had been solved completely by means of an algebraic calculus of “routes”, and remarked that the result was “of considerable practical importance”.

Despite an extensive search extending over several months, we have been unable to locate the original memoir. Bach and Bernul have themselves passed almost entirely from the historical record: according to a biographical note appended to a later railway report, Bach was killed in an accident upon the very line whose design had occupied his attention. No manuscript is known to survive, and the Hungarian notice remains the sole testimony to what must have been a remarkable piece of work. In the subsections that follow we reconstruct the essential argument from the information preserved in the notice, supplementing it where necessary with arguments of our own.

¹*Mathematikai és Fizikai Lapok* (Hungarian Mathematical and Physical Journal), Vol. V (1906), pp. 234–235. The notice reviews a paper by V. D. Z. Bach and A. M. Bernul entitled *Optimírte Layouts für den Spurwechsel*, published the previous year in the *Mittheilungen der Eisenbahnfreunde-Vereinigung Kopenhagen*, Vol. II, pp. 12–29.

4.6.1 Statement of the Problem

Throughout this section it is assumed that the junction is operated under a *fixed-direction regime*: every train entering the junction travels in one and the same direction—let us say, from west to east—and no train is ever required to reverse. This is the standard operational assumption for mainline junctions of the period, where running movements were strictly separated from shunting; it is also the assumption that gives the switch problem its mathematical bite, for it means that a switch permitting lateral passage from track a to track b conveys a train *while it continues forward*, and says nothing whatever about whether a train already on track b may return to a by the same piece of trackwork.

With this understanding in place, consider a junction consisting of n parallel tracks arranged in a fixed linear order $a_1, a_2, \dots, a_n$. A train enters upon one track and wishes to depart upon a different track, continuing throughout in the forward direction; no reversal of motion is permitted at any stage of the journey. The junction is equipped with *switches*, each of which is unidirectional in the lateral sense: a switch from track a to track b permits lateral passage in that direction only (that is, while the train proceeds forward), and implies nothing whatever about the possibility of lateral passage in the reverse direction from b to a . Moreover, since the tracks run in parallel, a switch may connect only two adjacent tracks in the linear order; a train cannot cross directly from a_i to a_j when $|i - j| > 1$.

A junction is called *perfect* if, for every ordered pair of distinct tracks (a_i, a_j) , there exists a sequence of switches by which a train entering on a_i can reach a_j without reversal. The problem is to determine whether perfect junctions exist, and, among those that do, to find one requiring the fewest switches.

4.6.2 The Algebraic Calculus of Routes

The Hungarian notice preserves a substantial portion of the notation employed by Bach and Bernul. We adopt their symbols, supplementing them where necessary for clarity.

A single switch leading from track a to an adjacent track b is written

$$a \mathcal{R} b.$$

A train that remains on its own track without moving through any switch is described by the *identity* $a \mathcal{R} a$, which is assumed to hold for every track a without requiring any physical installation. The first formal axiom of the calculus states that two switches are equal precisely when they connect the same pair of tracks in the same order:

$$a \mathcal{R} b = c \mathcal{R} d \iff a = c \text{ and } b = d.$$

In particular, no switch is its own reverse: $a \mathcal{R} b$ and $b \mathcal{R} a$ are distinct objects, and the existence of one in a given junction does not entail the existence of the other. This directional asymmetry is a direct consequence of the fixed-direction regime. Because all trains travel forward through the junction, the piece of trackwork that diverts a train from a onto b (say, to the left) is physically quite separate from the one that diverts it from b onto a (to the right); each must be designed, installed, and maintained independently. This is illustrated in Figures 4.1 and 4.2, which depict respectively a switch from a to b and the reverse switch from b to a .

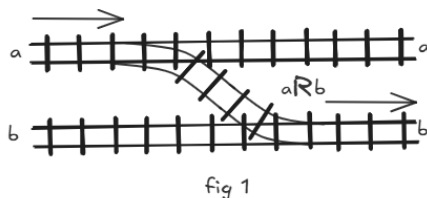


FIGURE 4.1. The switch $a \mathcal{R} b$: a train passes from track a to the adjacent track b .

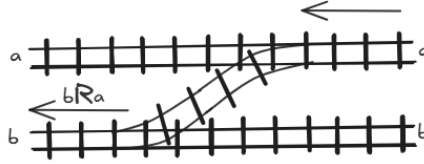


fig 2

FIGURE 4.2. The reverse switch bRa : lateral passage in the opposite direction, from b to a , while the train continues forward. Note that this is an independent installation, quite separate from Figure 4.1.

Were one to install a switch in every permitted lateral direction between every adjacent pair of tracks, the total would be $2(n - 1)$ switches. If, further, one were to allow switches between *all* pairs of distinct tracks (not merely adjacent ones), the count would rise to $n(n - 1)$. This latter figure serves as a crude upper bound:

$$\text{number of switches} \leq n(n - 1). \quad (4.1)$$

The problem is to determine how far below this bound one may descend while preserving the property of perfection.

To describe a multi-switch journey, Bach and Bernul introduce a *composition* operator, written $[+]$. When a train travels from a to b through one switch and then continues from b to c through a second, the combined journey is expressed as

$$aRb [+] bRc = aRc.$$

Composition is associative: the grouping of intermediate steps has no effect on the overall result. It is not, however, commutative: the expression $aRb [+] bRc$ describes a journey from a to c , whereas $bRc [+] aRb$ is meaningful only if $c = a$, since the second switch cannot be engaged until the train has already reached b . The physical ordering of switches along a route is therefore built directly into the algebraic structure.

For the purpose of recording which intermediate tracks a journey has visited, the calculus annotates the relation symbol by placing

a set of track-labels above it:

$$a \overset{S}{\mathcal{R}} b.$$

This expression denotes a path from a to b that has passed through each track in the set S , and through no other intermediate track. When $S = \emptyset$ the annotation is suppressed and one writes simply $a\mathcal{R}b$, as before. Composition of annotated paths is then defined by

$$a \overset{S_1}{\mathcal{R}} b \text{ } [+] \text{ } b \overset{S_2}{\mathcal{R}} c = a \overset{S_1 \cup \{b\} \cup S_2}{\mathcal{R}} c, \quad (4.2)$$

which records the junction track b as an intermediate stop and absorbs the complete histories of both constituent journeys into a single expression.

We now turn to a more general treatment of compound journeys involving several switches. When one wishes to analyse the total connectivity afforded by a set of installed switches—rather than the course of a single predetermined journey—it is convenient to write the available switches as a formal sum

$$r_1 \text{ } [+] \text{ } r_2 \text{ } [+] \text{ } r_3 \text{ } [+] \text{ } \cdots \text{ } [+] \text{ } r_n,$$

where each r_k stands for some switch $a\mathcal{R}b$. In this context the operator $[+]$ denotes the *union* of available routes rather than their sequential composition, and its terms may therefore be freely reordered: the set of switches installed in a junction is independent of the order in which one chooses to list them. In particular, for any index $i > 1$, the expression above may be rearranged as

$$r_1 \text{ } [+] \text{ } r_i \text{ } [+] \text{ } r_2 \text{ } [+] \text{ } \cdots \text{ } [+] \text{ } r_{i-1} \text{ } [+] \text{ } r_{i+1} \text{ } [+] \text{ } \cdots \text{ } [+] \text{ } r_n$$

without altering which journeys the junction permits. The value of this observation is that, once the switches have been collected into a union, one may scan them in any convenient order when searching for composable pairs—that is, pairs $r_j = a\mathcal{R}b$ and $r_k = b\mathcal{R}c$ whose endpoints match, and which therefore compose to give a longer

route $a\mathcal{R}c$. The physical arrangement of the junction is illustrated in Figure 4.3, which shows three tracks a , b , c equipped with the four switches²

$$a\mathcal{R}b, \quad c\mathcal{R}b, \quad b\mathcal{R}a, \quad b\mathcal{R}c.$$

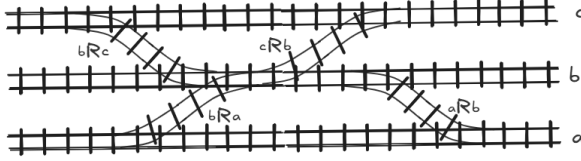


fig 3

FIGURE 4.3. A three-track junction equipped with four switches: $a\mathcal{R}b$, $c\mathcal{R}b$, $b\mathcal{R}a$, and $b\mathcal{R}c$. Reading the switches in either of the two orderings described in the text, one identifies the composed routes $a\mathcal{R}c$ and $c\mathcal{R}a$.

Written as a union, this junction is $a\mathcal{R}b \text{ } [+]\text{ } c\mathcal{R}b \text{ } [+]\text{ } b\mathcal{R}a \text{ } [+]\text{ } b\mathcal{R}c$. Applying the reordering rule, we may bring $b\mathcal{R}a$ forward past $c\mathcal{R}b$ to obtain

$$a\mathcal{R}b \text{ } [+]\text{ } b\mathcal{R}a \text{ } [+]\text{ } c\mathcal{R}b \text{ } [+]\text{ } b\mathcal{R}c.$$

Scanning now for composable pairs, we find $a\mathcal{R}b$ followed by $b\mathcal{R}c$, which compose to $a\mathcal{R}c$; and $c\mathcal{R}b$ followed by $b\mathcal{R}a$, which compose to $c\mathcal{R}a$. Hence the four switches of Figure 4.3 suffice to carry a train from a to c and, equally, from c to a —even though no direct switch between a and c has been installed. In the sequel we shall use this annotated calculus to formulate a precise account of which tracks a given route visits, a facility that proves especially convenient when verifying perfection.

4.6.3 Graphs and the Minimum Switch Theorem

The algebraic calculus is well adapted to formal deduction, but a visual representation is also instructive. Given any set of installed

²Read the direction of travel from top to bottom and from right to left, following the arrow on each switch.

switches, one constructs a *directed graph* in which each track a_i is a vertex and each switch $a_i\mathcal{R}a_j$ is a directed edge from a_i to a_j . A journey from one track to another then corresponds precisely to a directed path in this graph, and the junction is perfect if the graph is *strongly connected*—that is, if there exists a directed path from every vertex to every other vertex.

Figure 4.4 displays a strongly connected directed graph on three tracks, confirming that a perfect three-track junction is achievable. Figure 4.5 shows a directed graph on the same vertex set with an equal number of edges, yet one that fails to be strongly connected: at least one track cannot be reached from another, despite the presence of several switches. The contrast between the two figures makes plain that the *arrangement* of switches matters quite as much as their number.

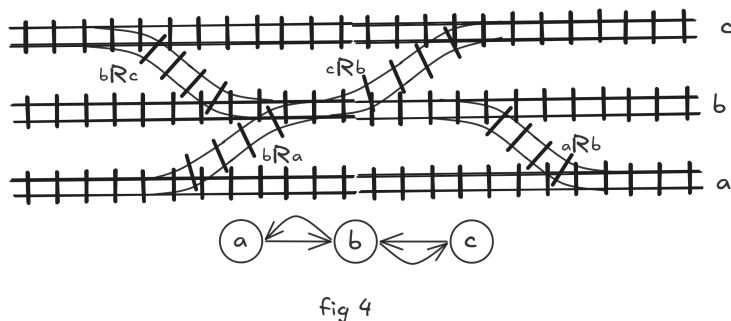


FIGURE 4.4. A strongly connected arrangement of switches on three tracks: every track is reachable from every other.

We are now in a position to state and prove the main result of this section.

Theorem 4.6.1. *The minimum number of switches required for a perfect n -track junction is $2(n - 1)$.*

Proof. We first establish that $2(n - 1)$ switches are necessary. The tracks $a_1, a_2, \dots, a_n$ are arranged in a fixed linear order, and a

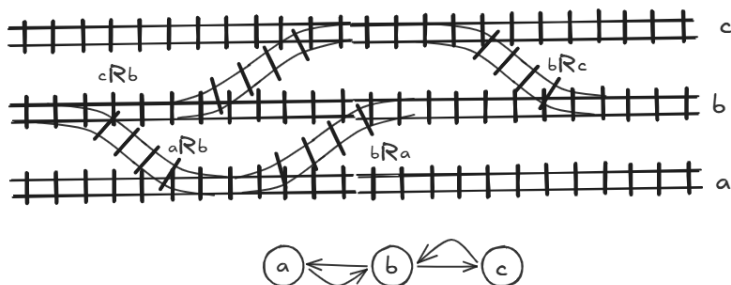


fig 5

FIGURE 4.5. An arrangement with the same number of switches as Figure 4.4, yet one that is not strongly connected. The mere presence of individual switches is no guarantee of complete reachability.

switch may connect only adjacent tracks. Consider the boundary between any two consecutive tracks a_i and a_{i+1} . In a perfect junction, a train must be able to travel from a_1 to a_n ; since the only route is along the chain, the boundary between a_i and a_{i+1} must be crossable in the forward lateral direction, so at least one switch $a_i\mathcal{R}a_{i+1}$ is required. By the same argument applied to the reverse lateral journey from a_n to a_1 , at least one switch $a_{i+1}\mathcal{R}a_i$ is required at the same boundary. These two switches are distinct unidirectional objects and neither implies the other. Since there are $n - 1$ adjacent boundaries, the total number of switches is at least $2(n - 1)$.

To show that this bound is attained, we exhibit a perfect junction containing exactly $2(n - 1)$ switches. For each $i = 1, 2, \dots, n - 1$, install one switch from a_i to a_{i+1} and one switch from a_{i+1} to a_i . This yields a symmetric chain

$$a_1 \Rightarrow a_2 \Rightarrow \dots \Rightarrow a_{n-1} \Rightarrow a_n,$$

in which any track a_i can reach any other track a_j by traversing the chain step by step in the appropriate lateral direction. The

junction is therefore perfect, and employs exactly $2(n-1)$ switches. $\square$

Combining Theorem 4.6.1 with the crude upper bound (4.1), we obtain the following double inequality for the number of switches in any perfect n -track junction:

$$2(n-1) \leq \# \text{ switches} \leq n(n-1).$$

For $n = 2$ the two bounds coincide, and the unique perfect two-track junction consists of exactly one switch in each lateral direction. For $n \geq 3$ the bounds diverge, and the question of which intermediate values are realizable by perfect junctions of various designs becomes a natural one. The Hungarian notice suggests that Bach and Bernul had begun to investigate this finer problem as well; their results on the matter, unhappily, have not survived.

A useful consequence of (4.2) is a compact condition for the existence of a connecting path. Suppose that the tracks $a_i, a_{i+1}, \dots, a_j$ are all connected in a perfect sub-junction with the associated n -relation (i.e. each adjacent pair is directly connected in both lateral directions), and that we introduce two further tracks a_{i-1} and a_{j+1} together with the relations $a_{i-1}\mathcal{R}a_i$ and $a_j\mathcal{R}a_{j+1}$. If the intermediate switches are written as the sum $\sum_{k=1}^n r_k$, then the composition

$$a_{i-1}\mathcal{R}a_i \left[+ \right] \left(\sum_{k=1}^n r_k \right) \left[+ \right] a_j\mathcal{R}a_{j+1}$$

yields the annotated relation

$$a_{i-1} \overset{S}{\mathcal{R}} a_{j+1},$$

where the set S contains every track a_x with $i \leq x \leq j$. The annotation S records exactly that history, confirming that no track inside the range is skipped, and in this way the annotated relation itself becomes a certificate of reachability across the entire block.